

Part B (Open Book)

Birla Institute of Technology and Science Pilani (Rajasthan)

Department of Electrical and Electronics Engineering

EEE F313/ INSTR F313, Analog and Digital VLSI Design

Second Semester

[Time: 60 Minutes]

Comprehensive Examination (Open Book)

Date: 15 May 2024

Max. Marks: 40

General Instructions to the candidate

- Symbols, constants, terminology used have their usual meaning unless specified specially.

Unless specifically mentioned—

Common data: Use the following common data (250nm technology) if not mentioned specifically in the question

Consider $V_{dd}=2.5\text{ V}$, $L_{min}=W_{min}=250\text{ nm}$, $R_{nmos}=13\text{ kohm/unit (W/L)}$, $R_{pmos}=31\text{ kohm/unit (W/L)}$,

$V_{TON}=0.4\text{ V}$, $V_{TOP}=-0.43\text{ V}$, $\lambda_n=0.1\text{ V}^{-1}$, $\lambda_p=0.1\text{ V}^{-1}$, $u_nC_{ox}=120\text{ uA/V}^2$, $u_pC_{ox}=40\text{ uA/V}^2$

Gate capacitance of minimum size MOSFET $C_{g,min}=0.05\text{ pico Farad}$

- Neglect body effect, channel length modulation if not mentioned specifically.
- The body of all PMOS is tied to V_{DD} while all NMOS is tied to ground unless specially mentioned.

Q1: The layout of a CMOS inverter is shown below. The first inverter is driving another inverter with the transistor widths three times larger maintaining the other dimensions same. Calculate the average propagation delay (τ_p) for the first inverter. The MOS parameters are given below:

$V_{dd}=5\text{ V}$, $V_{TON}=1\text{ V}$, $V_{TOP}=-1\text{ V}$, $k_n'=40\text{ uA/V}^2$, $k_p'=20\text{ uA/V}^2$, $C_{ox}=69\text{ nF/cm}^2$, $C_{jsw}=2.2\text{ pF/cm}$, $C_{jo}=7\text{ nF/cm}^2$, lateral overlap (L_D) = $1\text{ u}\mu\text{m}$, $L_{mask}=5\text{ u}\mu\text{m}$, the voltage equivalence factor for both n and p MOS are $K_{eq,n}=K_{eq,p}=0.436$. The source and drain region lengths are $Y=12\text{ u}\mu\text{m}$ and the channel width is $10\text{ u}\mu\text{m}$ for both transistors (NMOS and PMOS)

Assumptions: (i) interconnect capacitance between the inverter is negligible (ii) the junction capacitance for the area facing the channel side is negligible. (iii) input rise and fall times = 100 ps

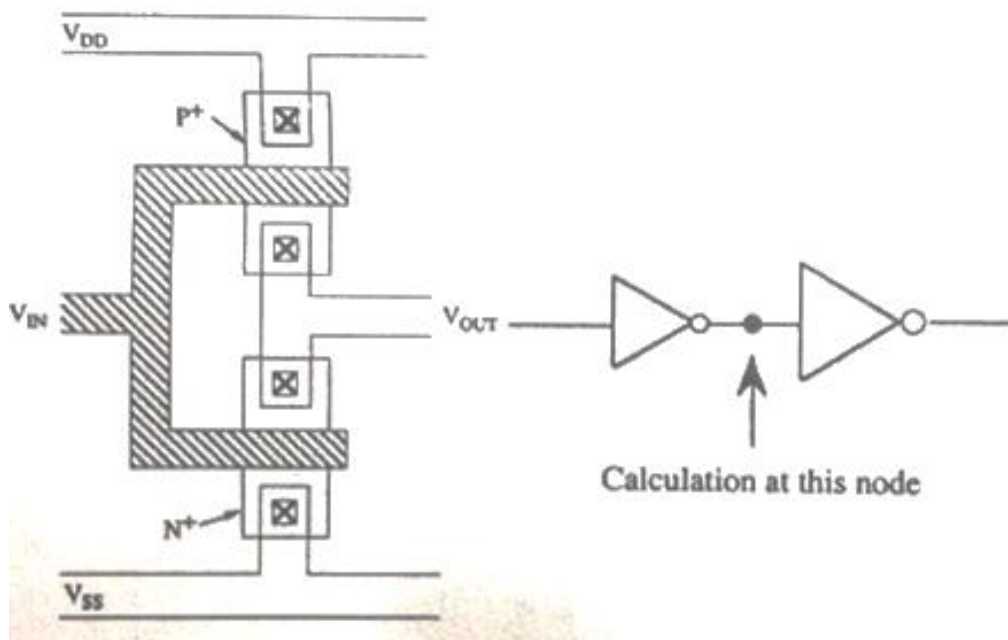


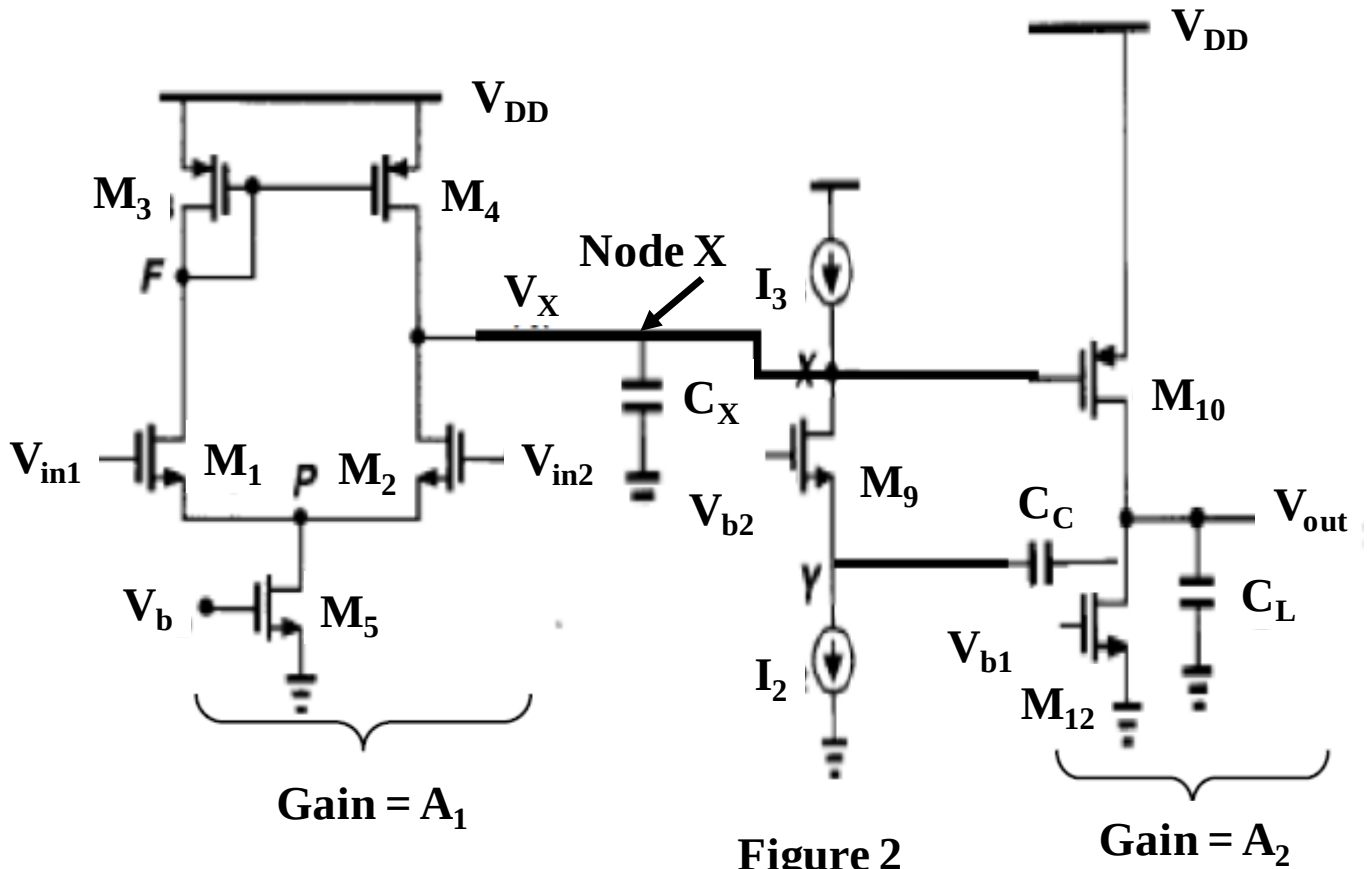
Figure 1

20 Marks

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Q2. For the circuit shown in Figure 2, the transistor pairs (M1, M2), (M3, M4) are matched and all transistors are operating in active region of operation. $V_{id}=V_{in1}-V_{in2}$. Assume DC current $I_5=2I_{12}=100\mu A$, $V_{ov.}=0.2V$. $C_L=C_c=1pF=3C_x$

20 Marks



- (a). Draw the small signal model
- (b). Derive the expressions (at low frequency) for:
 - i. Differential Mode Gain, $A_{DM} = [V_{out}/V_{id}]$,
 - ii. Common Mode Gain, $A_{CM} = [V_{out}/V_{CM}]$, and
 - iii. Common mode Rejection Ratio (CMRR) $= \frac{A_{DM}}{A_{CM}}$.
- (c). Prove that due to Miller effect, there is an additional capacitance appearing at node V_x which is equal to $|A_2 C_C|$
- (d). Intuitively, determine the value of dominant pole and zero frequencies of the entire amplifier

Note:

$$A_{CM} = \frac{V_{out}}{V_{CM}} = \frac{V_{out}}{\frac{V_{in1} + V_{in2}}{2}}$$
$$A_{DM} = \frac{V_{out}}{V_{DM}} = \frac{V_{out}}{V_{in1} - V_{in2}}$$

Comprehensive Examination (Closed Book)	Date: 15 May 2024	Max. Marks: 60
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General Instructions to the candidate

- Please write your answers legibly and neatly in the answer booklet provided only.
- Symbols, constants, terminology used have their usual meaning unless specified specially.
- There are a total of **TEN** questions, and all are compulsory. **Each question carries 6 marks.**

Unless specifically mentioned—

Common data: Use the following common data (250nm technology) if not mentioned specifically in the question

Take $V_{DD}=2.5\text{ V}$, $L_{min}=W_{min}=250\text{ nm}$, $R_{nmos}=13\text{ kohm/unit (W/L)}$, $R_{pmos}=31\text{ kohm/unit (W/L)}$,

$V_{TON}=0.4\text{ V}$, $V_{TOP}=-0.43\text{ V}$, $\lambda_n=0.1\text{ V}^{-1}$, $\lambda_p=0.1\text{ V}^{-1}$, $\mu_n C_{ox}=120\text{ uA/V}^2$, $\mu_p C_{ox}=40\text{ uA/V}^2$

Gate capacitance of minimum size MOSFET $C_{g,min}=0.05\text{ pico Farad}$

- Use **square law current equation** unless specifically mentioned. Use **lambda based design rules** unless specially mentioned.
- Neglect body effect, channel length modulation if not mentioned specifically.
- The body of all PMOS is tied to V_{DD} while all NMOS is tied to ground unless specially mentioned.

Q1. Write down the design steps involved in the fabrication of the device structure shown in Figure 1.

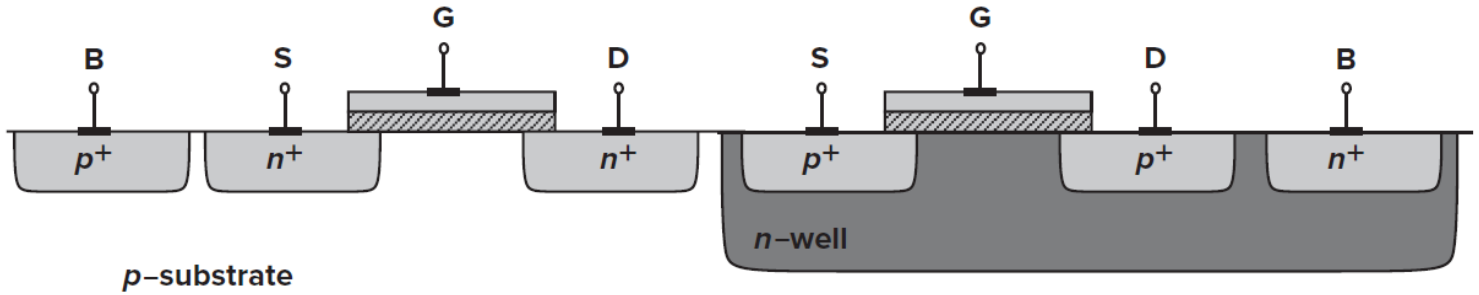


Figure 1

Q2. Draw symbolic representation of the device structure shown in Figure 2. Write down the conventional name of each terminal and the entire structure. Comment upon its region of operation if $V_G=1\text{ V}$ and $V_{TH}=0\text{ V}$.

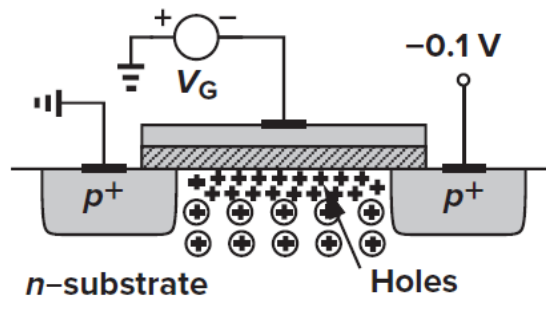


Figure 2

Q3. For the circuit shown in Figure 3, plot the on resistance as a function of the gate voltage (V_G). Consider, $\mu_n C_{ox} = 50 \mu\text{A}/\text{V}^2$, $W/L = 10$, and $V_{TH} = 0.3 \text{ V}$. Note that the drain terminal is open.

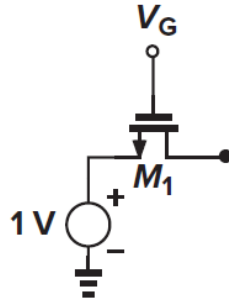


Figure 3

Q4. For the circuit shown in Figure 4, plot the trans-conductance (G_m) as a function of the voltage V_{DS} .

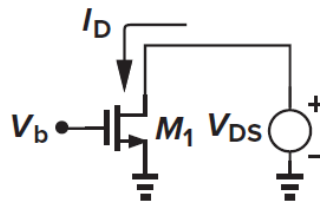


Figure 4

Q5. For the circuit shown in Figure 5, plot the drain current as V_X varies from $-\infty$ to 0. Assume $V_{TH0} = 0.3 \text{ V}$, Body effect coefficient, $\gamma = 0.4 \text{ V}^{1/2}$, and $2\phi_F = 0.7 \text{ V}$.

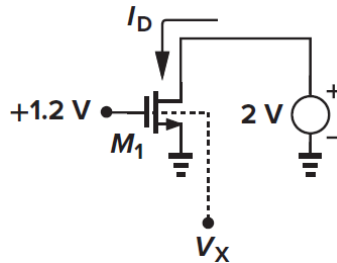


Figure 5

Q6. Write down the names and expressions for the different capacitances shown in Figure 6 in **Cutoff**, **Saturation** and **Linear** regions of operation.

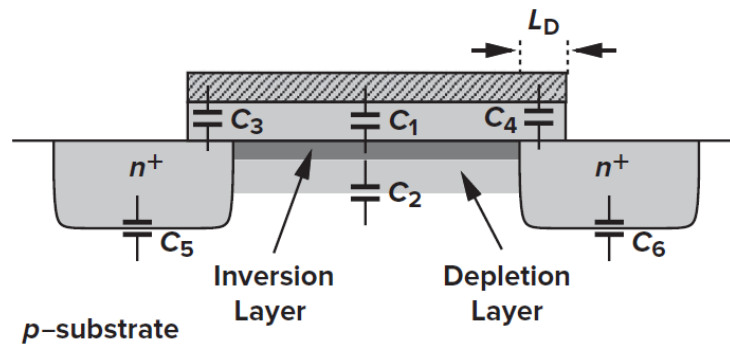


Figure 6

Q7. Calculate the source and drain junction capacitances of the two structures shown in Figure 7.

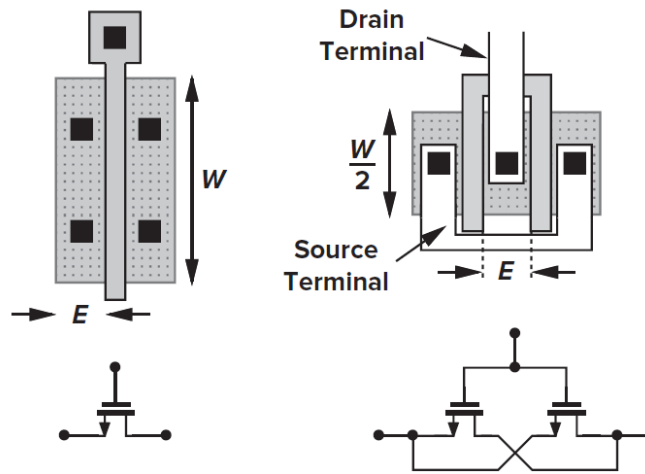


Figure 7

Q8. Synthesize a CMOS logic circuit that implements the Boolean Function Y. Also, mention the transistor sizing (W/L) for the pull-up and pull down devices. Consider $\left(\frac{W}{L}\right)_P = 2 \left(\frac{W}{L}\right)_N$ in a nominal CMOS inverter .

$$Y = \overline{A + B(C + D)}$$

Q9. For the inverter circuit shown in Figure 8, identify the topology of the inverter circuit and hence determine $\left(\frac{W}{L}\right)_N$ for the N-MOS Transistor (M_N) such that the low output voltage, $V_{OL} < 100$ mV. Consider, $\left(\frac{W}{L}\right)_P = \frac{3\mu\text{m}}{0.18\mu\text{m}}$, for the P-MOS Transistor, $\mu_n C_{ox} = 100 \mu\text{A/V}^2$, $\mu_p C_{ox} = 50 \mu\text{A/V}^2$, $V_{TON} = 0.4\text{V}$, $V_{TOP} = -0.5\text{V}$. Ignore channel length modulation.

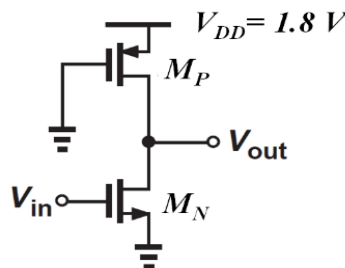


Figure 8

Q10. Sketch the capacitances, C_{EtoF} , C_{EtoN} , $C_{FtoBODY}$, $C_{NtoBODY}$ of the transistor M_1 in Figure 9, as V_X varies from zero to 3 V. Assume that $V_{TH} = 0.3\text{V}$ and $\lambda = \gamma = 0$

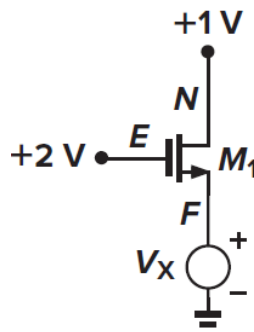


Figure 9